

#This code will be used to demonstrate theorem 6.18 • We will demonstrate that Q2

$$= \frac{((w+z)^3 - y^3) \cdot x \cdot (-w^2 \cdot (z^3 - y^3) + z^2 \cdot (w^3 - y^3))}{y \cdot (w+z)^2 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3))} \text{ is increasing along the } \gamma_2 \text{ curve; (Part 2 - discarding potential critical points)}$$

#Define r2

$$\begin{aligned} r2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + \right. \\ & -4 b^{18} + 8 b^{16} - 4 b^{14} \Big) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \\ & + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \Big) b^8 a^{24} - 4 b^2 (b^{28} \\ & + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} \\ & + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} \\ & - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 \\ & - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} \right. \\ & + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \Big) b^6 a^{22} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 b^6 \left(b^{20} + b^{18} \right. \\ & - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \Big) \\ & a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\ & - 27212 b^8 - 6620 b^6 - 2636 b^4 + 36 b^2 + 1) a^{12} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16} \\ & - 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18} \\ & - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} + 8 b^{30} \\ & + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\ & - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-9 b^{24} \\ & - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\ & + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\ & - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\ & - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \end{aligned}$$

$$\begin{aligned}
& + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\
& + 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \\
& \left. + \frac{8}{5} b^2 - 1 \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big):
\end{aligned}$$

#The *candidate points*· (a, b) for the intersection of $r_2 = 0$ and $\tilde{f} = 0$ are of the form $a \in A$ and $b \in B$ where

$A := [1.839289820, 3.474957787, 3.883374171, 3.888068164] :$

$B := [1.345726290, 1.629433656] :$

$Points := [seq(seq([a, b], b \text{ in } B), a \text{ in } A)] :$

$Points_filtered := select(p \rightarrow p[2] < p[1], Points);$

$Points_filtered := [[1.839289820, 1.345726290], [1.839289820, 1.629433656], [3.474957787, 1.345726290], [3.474957787, 1.629433656], [3.883374171, 1.345726290], [3.883374171, 1.629433656], [3.888068164, 1.345726290], [3.888068164, 1.629433656]]$ (1)

1.066555212, 1.304932132

$Points_filtered := [[3.192177604, 1.748533112], [3.538902695, 1.748533112], [3.538902695, 3.464457702], [3.555974116, 1.748533112], [3.555974116, 3.464457702], [3.643966007, 1.748533112], [3.643966007, 3.464457702], [3.884436541, 1.748533112], [3.884436541, 3.464457702]]$ (2)

with(plots) :

with(plottools) :

$r := a - b :$

$p1 := \text{implicitplot}([r_2 = 0, r = 0],$
 $a = 1 \dots 4, b = 1 \dots \text{sqrt}(3),$
 $\text{scaling} = \text{constrained},$
 $\text{gridrefine} = 2,$
 $\text{thickness} = 2,$
 $\text{color} = [\text{blue}, \text{red}],$
 $\text{labels} = ["a", "b"],$
 $\text{title} = \text{"Graphic for } r_2=0"$
 $) :$

retangles

$R1 := \text{rectangle}([1.5, 1], [2, 1.4],$
 $\text{color} = \text{silver}, \text{transparency} = 0.5,$
 $\text{linestyle} = 3, \text{ # pontilhado}$

```

        thickness = 2 ) :
R2 := rectangle( [ 1.839, 1.629 ], [ 1.840, 1.630 ],
        color = silver, transparency = 0.5,
        linestyle = 3, # pontilhado
        thickness = 2 ) :
R3 := rectangle( [ 3, 1.3 ], [ 4, 1.7 ],
        color = silver, transparency = 0.5,
        linestyle = 3, # pontilhado
        thickness = 2 ) :

```

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# points

```

```

P := pointplot(
    Points_filtered,
    symbol = solidcircle,
    symbolsize = 12,
    color = black
) :

```

```

# sobrepor tudo

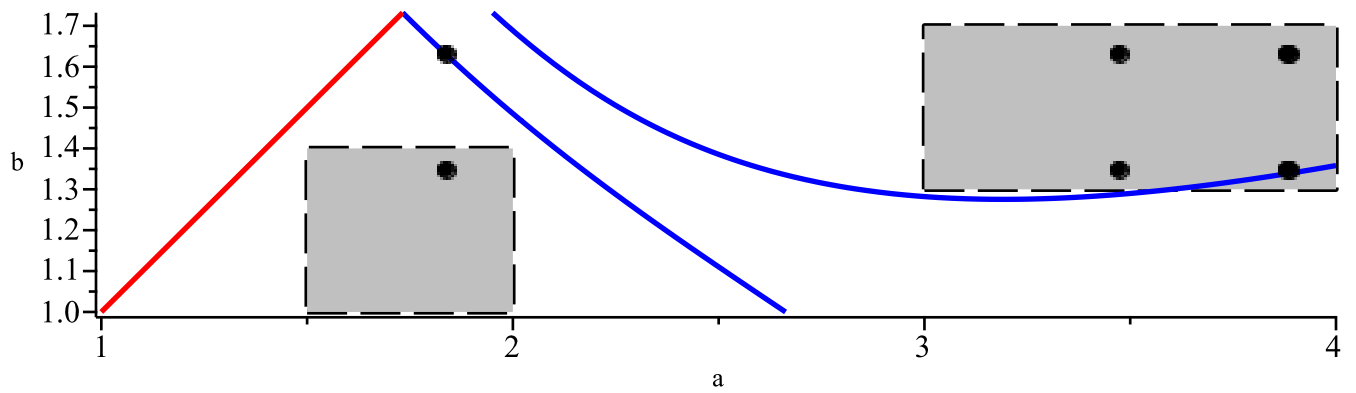
```

```

display( [p1, P, R1, R2, R3] );

```

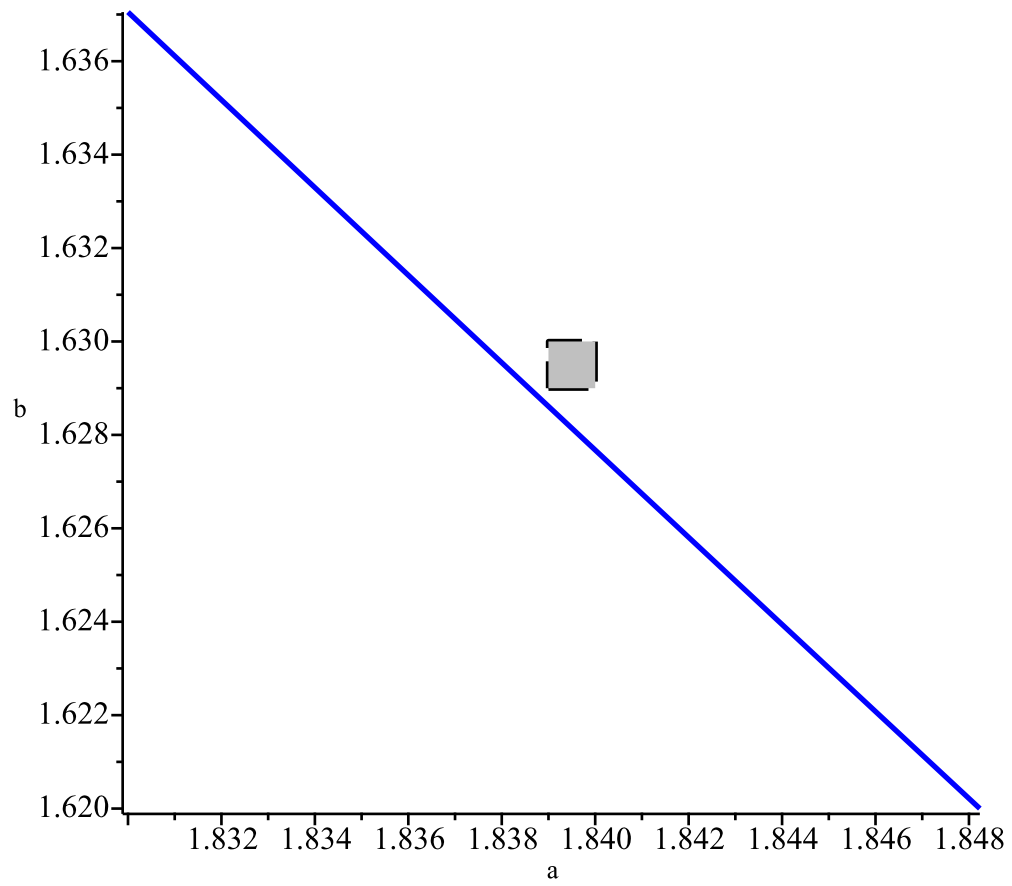
Graphic for $r_2=0$



```
p2 := implicitplot(
[r2 = 0, r = 0],
a = 1.83 .. 1.93, b = 1.62 .. 1.72,
scaling = constrained,
gridrefine = 2,
thickness = 2,
color = [blue, red],
labels = ["a", "b"],
title = "Graphic for r2=0"
):
```

```
display([p2, R2]);
```

Graphic for $r_2=0$



restart;

#Define r2

$$\begin{aligned} r2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + \right. \\ & -4 b^{18} + 8 b^{16} - 4 b^{14} \left. \right) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \\ & + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \left. \right) b^8 a^{24} - 4 b^2 (b^{28} \\ & + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} \\ & + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} \\ & - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 \\ & - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} \right. \\ & + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 b^6 \left(b^{20} + b^{18} \right. \\ & - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \left. \right) \\ & a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\ & - 27212 b^8 - 6620 b^6 - 2636 b^4 + 36 b^2 + 1) a^{12} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16} \\ & - 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18} \\ & - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} + 8 b^{30} \\ & + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\ & - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-9 b^{24} \\ & - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\ & + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\ & - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\ & - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \\ & + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \end{aligned}$$

$$+ 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 + \frac{8}{5} b^2 - 1 \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big):$$

#Apply the Möbius transformation on the rectangle $(1.5, 2) \times (1, 1.4)$ that contains one point, constructing r^2

$$moebius := \left\{ \begin{array}{l} a = \frac{\frac{3}{2} + 2 \cdot c}{1 + c}, \\ b = \frac{1 + \frac{7}{5} \cdot d}{1 + d} \end{array} \right\} :$$

```
r2 := collect(numer(expand(subs(moebius, r2))), c) :
```

#Compute the degree of r^2 in c

$$degree(r2, c);$$

32

(3)

#Define the coefficients $c_i = \text{coeff}(r^2, c, i)$

for i from 0 to 32 do
$$c[i] := \text{coeff}(\text{'r2'}, c, i);$$

od:

#Compute the all degree of c_i in d

```
seq( degree( c[i], d ), i = 0 .. 32 );
```

[illegible]

#From here we will analyze the signal variation of all coefficients c_i

$$seq(seq(sign(coeff(c[j], d, i)), i = 32..0, -1), j = 32..0, -1);$$
[illegible]

[illegible]

[illegible]

restart;

#Define r2

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[illegible]

[illegible]